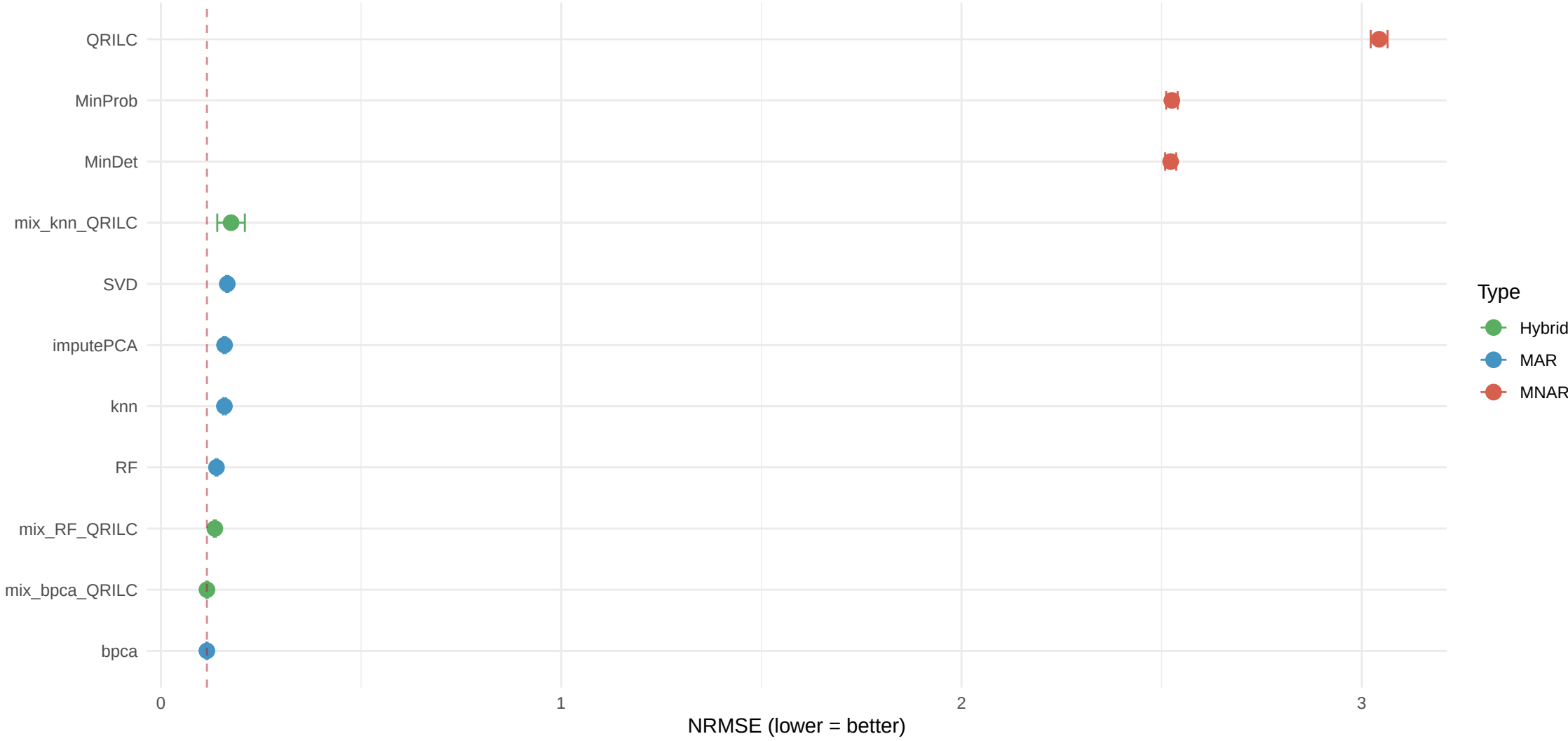


Imputation Benchmark

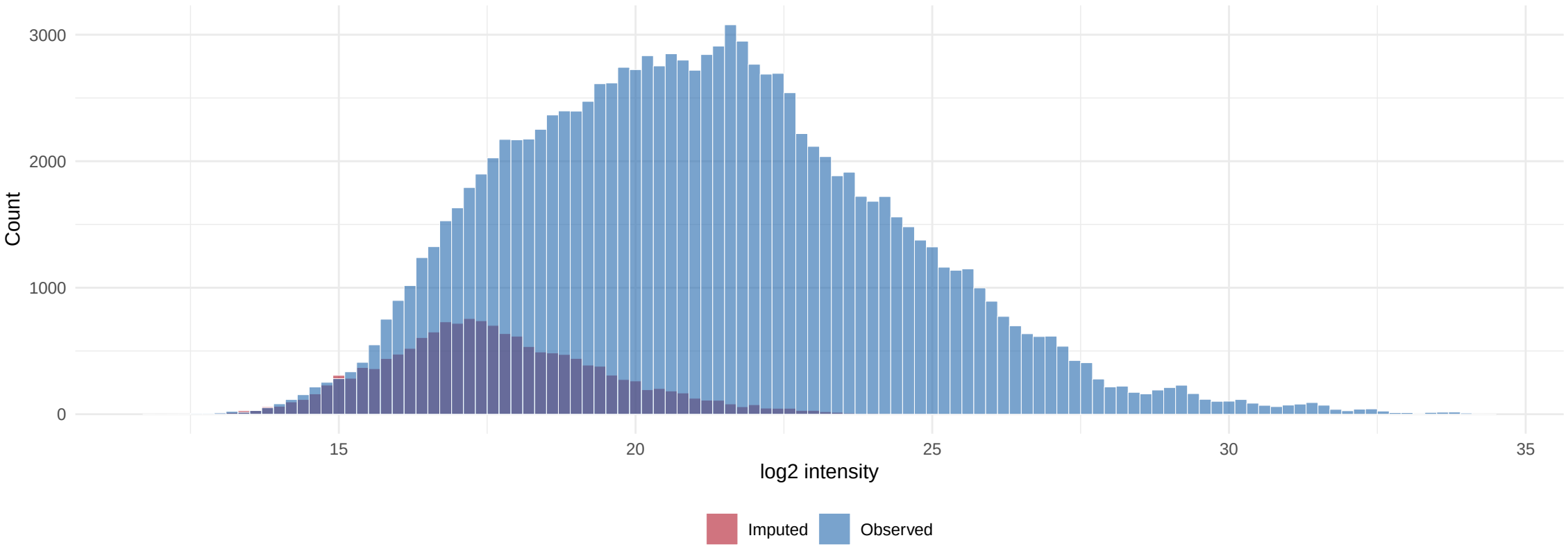
Imputation Benchmark
20 iter x 10% masked | Best: bpca (0.1147) | Off-scale: zero



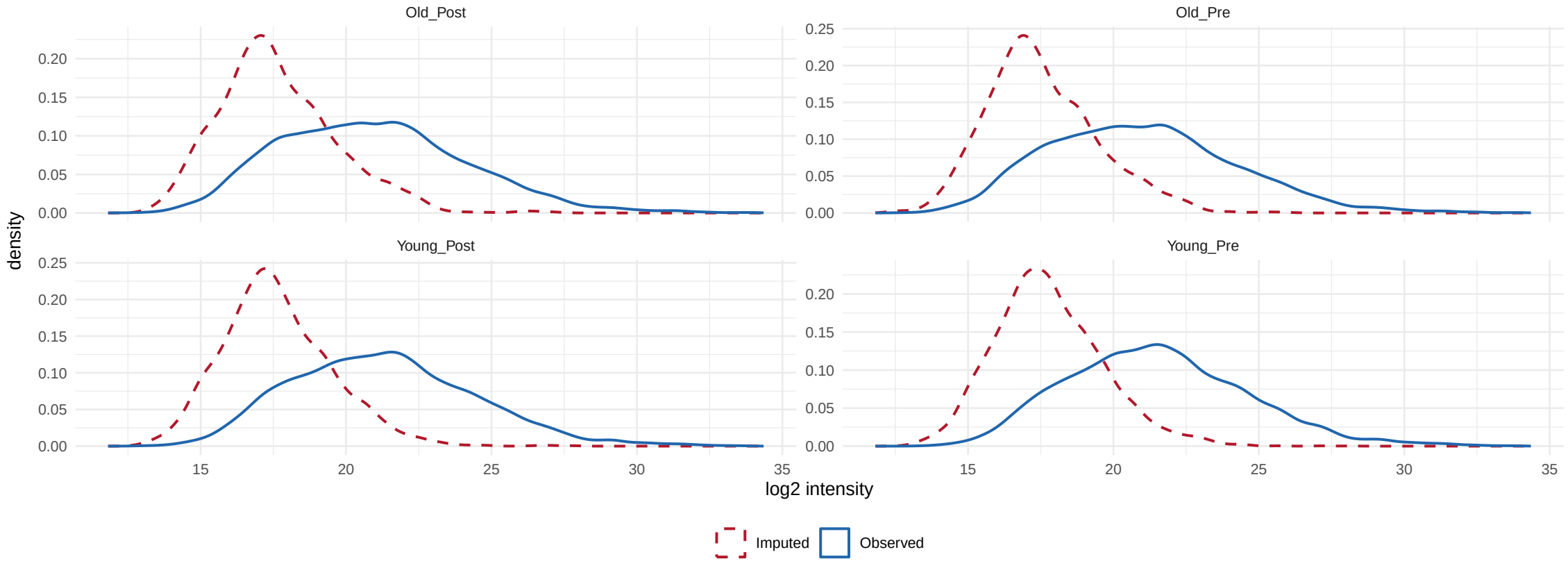
PCA Structure Preservation
Lower PSS = less geometric distortion of sample relationships



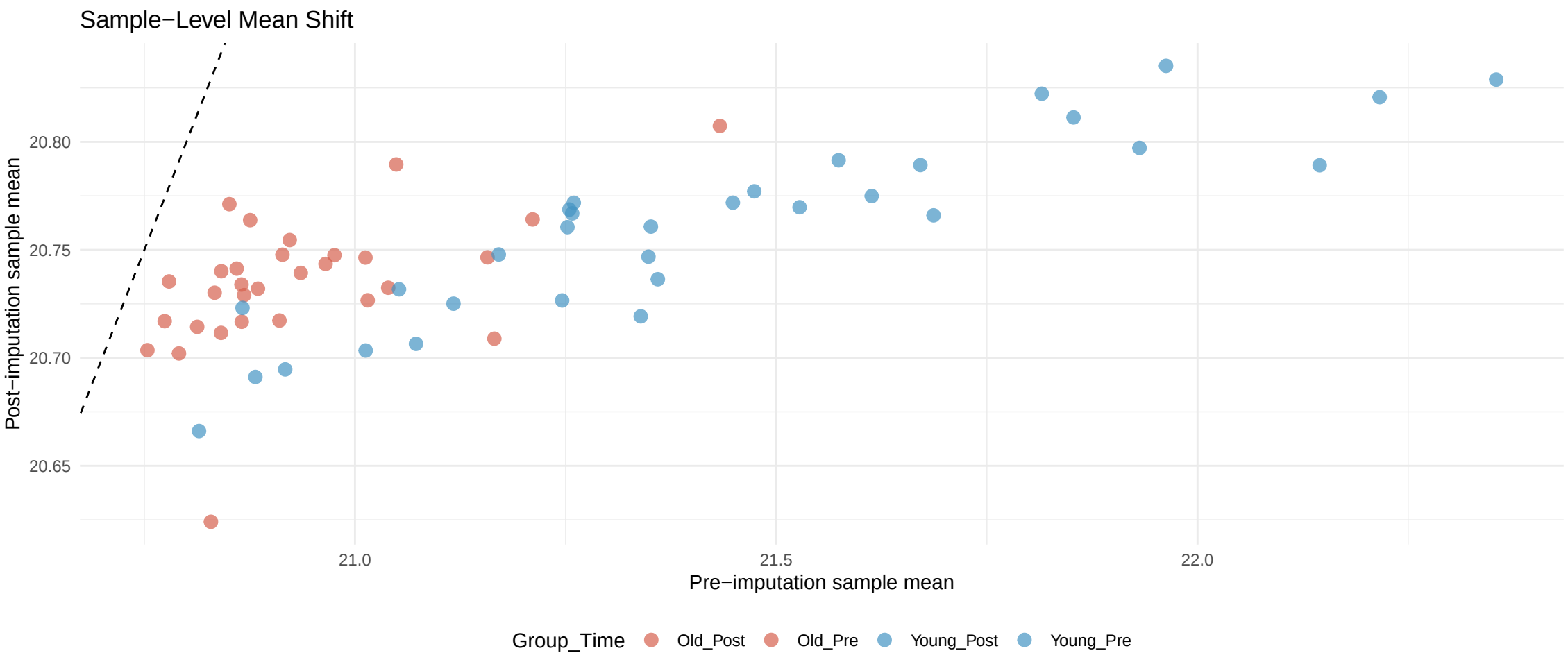
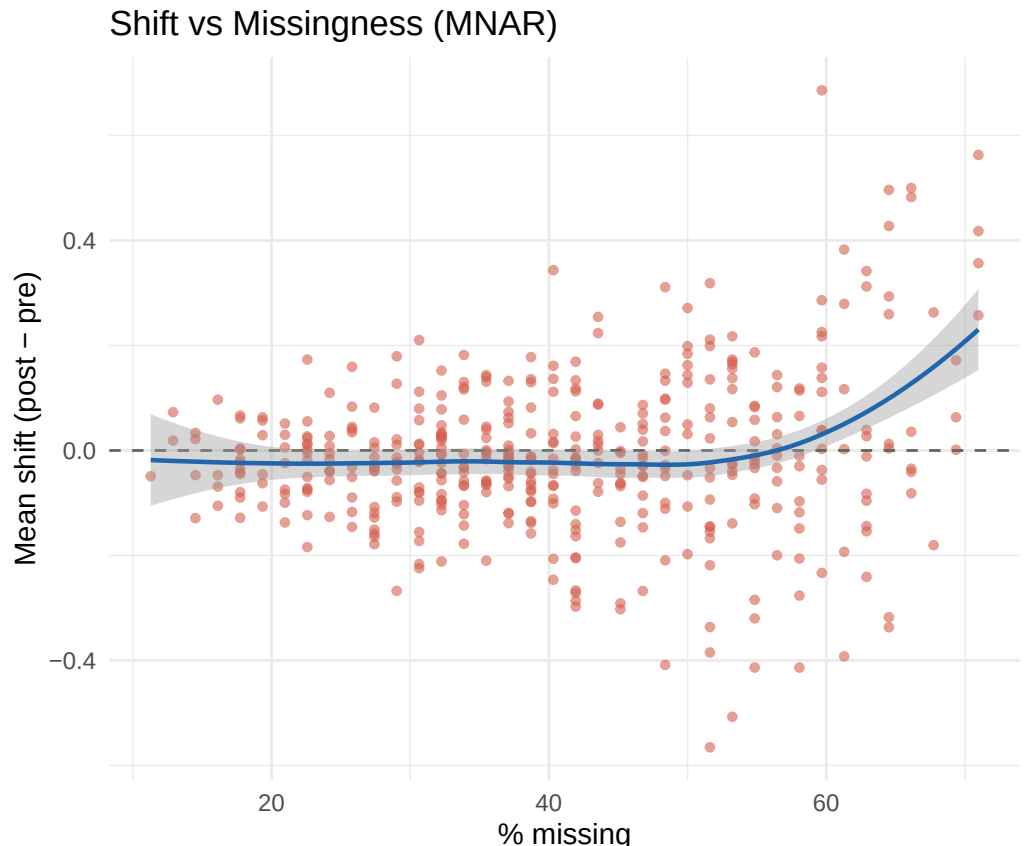
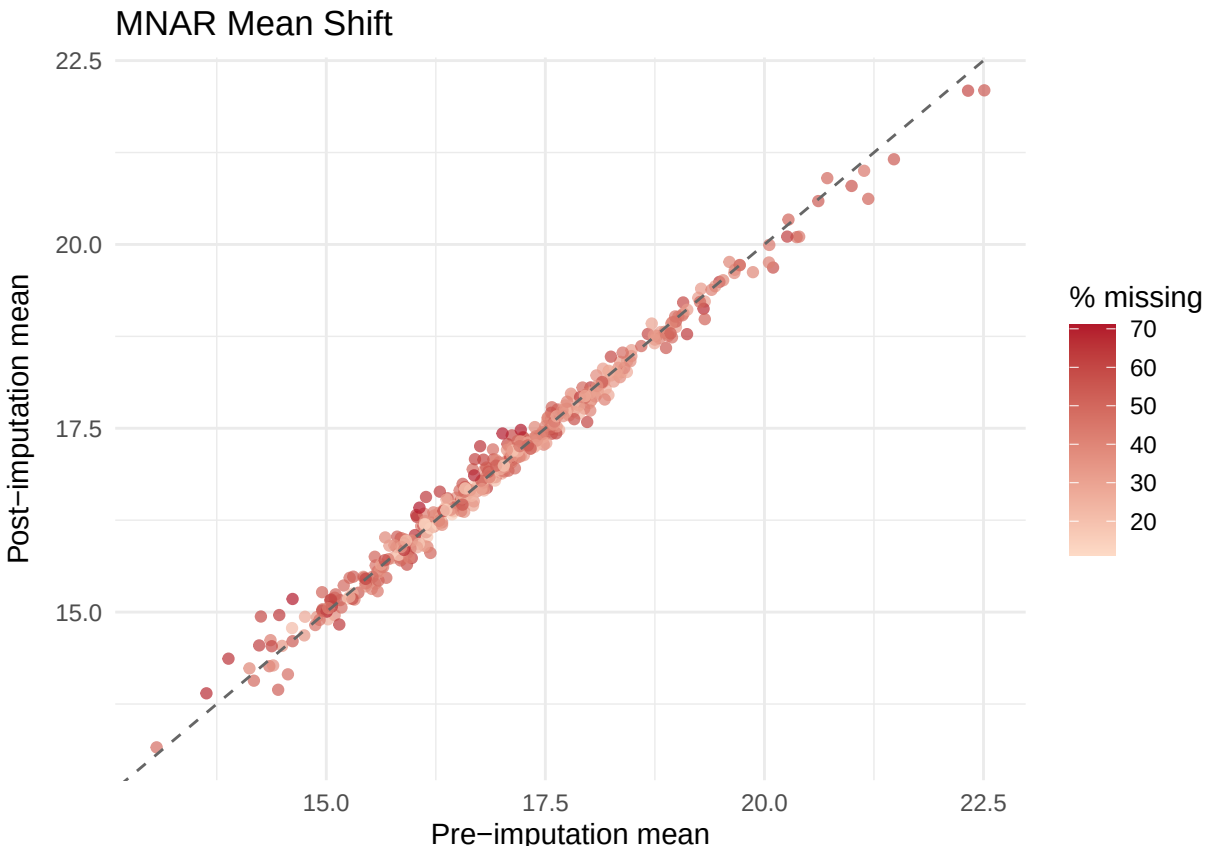
Observed vs Imputed Distributions
116,317 observed | 15,371 imputed



Observed vs Imputed by Group

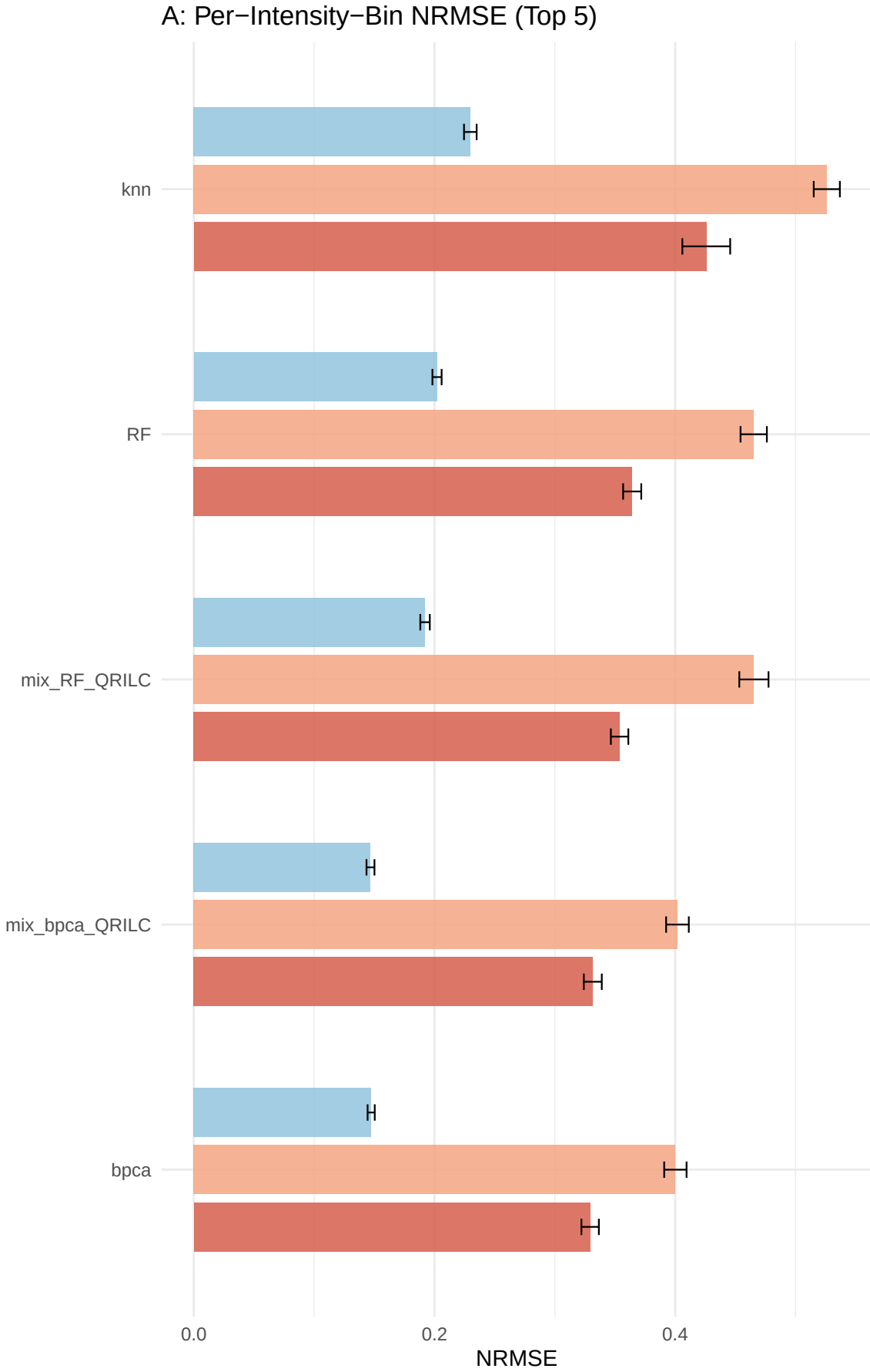


MNAR Audit (432 proteins) & Sample Shift



Extended Diagnostics (Top 5 Methods)

Per-intensity accuracy and method summary



B: Extended Summary

Intensity tertile	method	nrmse	pss	nrmse_high	nrmse_low	nrmse_mid
	bpca	0.1147	26.4556	0.1474	0.3293	0.4001
	mix_bpca_QRILC	0.1150	25.3919	0.1468	0.3315	0.4019
	mix_RF_QRILC	0.1345	32.4168	0.1921	0.3538	0.4654
	RF	0.1381	32.2436	0.2021	0.3643	0.4652
	knn	0.1581	43.1513	0.2298	0.4258	0.5259

Per-Sample Imputation Accuracy

Per-Sample LOOCV (bpca, 20 iter × 10% mask)
High/Low mean NRMSE ratio: 1.48 | Ref line: bpca benchmark (0.1147)

